THANOS : Enhancing Conversational Agents with *Skill-of-Mind*-Infused Large Language Model

Young-Jun Lee [1] **Dokyong Lee** [2] **Junyoung Youn** [2] **Kyeongjin Oh** [2] **Ho-Jin Choi** [1]
[1] School of Computing, KAIST [2] KT Corporation
{yj2961, hojinc}@kaist.ac.kr {dokyong.lee, junyoung.youn, kyeong-jin.oh}@kt.com

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Abstract

To increase social bonding with interlocutors, humans naturally acquire the ability to respond appropriately in a given situation by considering which conversational skill is most suitable for the response — a process we call *skill-of-mind* [1]. For large language model (LLM)-based conversational agents, planning appropriate conversational skills, as humans do, is challenging due to the complexity of social dialogue, especially in interactive scenarios. To address this, we propose a *skill-of-mind*-annotated conversation dataset, named MULTIFACETED SKILL-OF-MIND, which includes multi-turn and multifaceted conversational skills across various interactive scenarios (e.g., long-term, counseling, task-oriented), grounded in diverse social contexts (e.g., demographics, persona, rules of thumb). This dataset consists of roughly 100K conversations. Using this dataset, we introduce a new family of *skill-of-mind*-infused LLMs, named THANOS , with model sizes of 1B, 3B, and 8B parameters. With extensive experiments, these models successfully demonstrate the *skill-of-mind* process and exhibit strong generalizability in inferring multifaceted skills across a variety of domains. Moreover, we show that THANOS significantly enhances the quality of responses generated by LLM-based conversational agents and promotes prosocial behavior in human evaluations. Code: https://github.com/passing2961/Thanos.

1 Introduction

In everyday conversations, humans engage in diverse and complex interactions with their interlocutors (e.g., friends, colleagues) by understanding and interpreting their interlocutors' situations (Rashkin, 2018; Lee et al., 2022a) and personas (Zhang, 2018; Lee et al., 2022b), and by recalling memorable events or moments (Bae et al., 2022a; Jang et al., 2023; Lee et al., 2024d). Humans do not always know the most appropriate response for every turn in a multi-turn conversation. Instead, they learn to choose suitable responses by internally considering the most appropriate conversational skills at that moment, based on the interlocutor and social context over time. As illustrated in Figure 1, for example, humans reflect on which skill to use for the next turn by internally reasoning about which skill would be most appropriate. This process evolves through self-reflection and feedback, as people assess the positive or negative reactions of their interlocutors. We refer to this entire process as *skill-of-mind*, which involves interpreting and understanding the current dialogue situation, planning the best

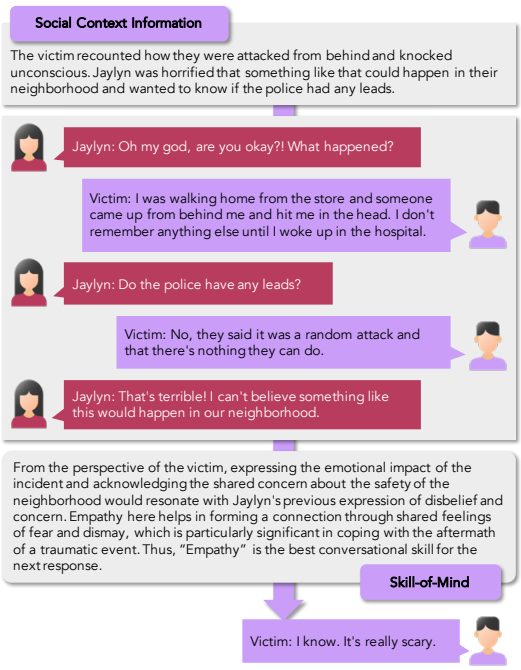

Figure 1: An overview of *skill-of-mind* process.

[1]We derive the term *skill-of-mind* from *theory of mind* (Premack and Woodruff, 1978), which refers to the ability to understand and infer others' mental states, intentions, and beliefs from situational descriptions (e.g., narratives). Skill-of-mind refers to interpreting/understanding the current conversational context based on social dynamics (e.g., demographics, persona) and optimizing social interaction through conversational skills.

skill strategy for the next response, and then selecting the most appropriate conversational skill.

Recently, conversational agents powered by LLMs (Touvron et al., 2023; AI@Meta, 2024; Team et al., 2024) have demonstrated impressive capabilities in logical reasoning (Pan et al., 2023) and creativity (Franceschelli and Musolesi, 2023). However, these agents still struggle with social commonsense reasoning (Chae et al., 2023) and strategic communication skills in an interactive environment (Zhou et al., 2023). Additionally, despite interacting with hundreds of millions of individual users (Zhao et al., 2024), each possessing a distinct perspective or persona, current agents do not effectively personalize their responses to users (Lee et al., 2024b). We posit that directly generating the next response is particularly challenging for LLM-based conversational agents due to the complexity of social dialogue, specifically: (1) the *one-to-many* problem (Li et al., 2015; Bao et al., 2019), where multiple plausible responses exist, and (2) the scattering of key evidence (Chae et al., 2023), where critical information or phrases are dispersed throughout the conversation. To address this, we suggest that before generating the next response, interpreting the dialogue situation and planning the most plausible conversational skill — similar to how humans' internal mind states operate — can enhance response quality, even in interactive situations involving social dynamics, by acting as a form of guidance.

To this end, we introduce MULTIFACETED SKILL-OF-MIND, a collection of multi-turn, multifaceted *skill-of-mind* annotated conversations. This dataset includes annotations for both explanations and conversational skills, covering a wide range of conversational skills across one-sided turns within dialogues. The dataset is derived from 12 existing source dialogue datasets, which encompass diverse social contexts and scenarios (e.g., chitchat, counseling). To annotate *skill-of-mind*, we prompt GPT-4 (Achiam et al., 2023) (i.e., `gpt-4-turbo`) to generate explanations and identify conversational skills from a predefined collection of conversational skills, organized hierarchically into five main categories: Interpersonal, Memory & Knowledge Management, Cognitive & Problem-Solving, Communication & Listening, Task-Oriented.

Additionally, we propose a new family of *skill-of-mind*-infused LLMs, THANOS , which generate both an explanation and the most appropriate conversational skill in LLM-based conversational agents. Through extensive experiments, THANOS accurately predicts conversational skills and generates high-quality explanations across various dialogue scenarios, demonstrating strong *generalizability* in skill prediction. To further validate the effectiveness of THANOS , we incorporate the generated *skill-of-mind* as augmented input prompts for LLM-based conversational agents when responding to the next turn, resulting in significant improvements in response quality.

Our contributions are summarized as follows: (1) We introduce a new social concept, *skill-of-mind*, which involves interpreting dialogue situations, planning the best skill strategy, and selecting the appropriate conversational skill. (2) We present a multi-turn, multifaceted *skill-of-mind*-annotated conversation dataset, MULTIFACETED SKILL-OF-MIND, which encompasses diverse social dynamics and interactive scenarios. (3) Using MULTIFACETED SKILL-OF-MIND, we propose a family of *skill-of-mind*-infused LLMs, THANOS , with model sizes of 1B, 3B, and 8B parameters. With extensive experiments, we demonstrate the effectiveness and generalizability of THANOS across various scenarios.

2 MULTIFACETED SKILL-OF-MIND

In this section, we explain the definition of *skill-of-mind* (§ 2.1), why predicting *skill-of-mind* is necessary (§ 2.2), the hierarchical taxonomy of conversational skills (§ 2.3), and the data construction process (§ 2.4). Lastly, we analyze the constructed dataset (§ 2.5).

2.1 Definition of "Skill-of-Mind"

We begin by defining what "skill-of-mind" means in the context of this work. In the conversational AI literature (Smith, 2020; Kim et al., 2022c), these "skills" are sometimes regarded as communication strategies (Zhou et al., 2023), but generally refer to the desirable abilities required to maintain continuous and meaningful conversations with an interlocutor. These skills cover a broad spectrum, ranging from general abilities (e.g., empathy, persona) to task-specific functions (e.g., making phone calls, booking a hotel). In everyday social interactions, which are a core component of human life, people typically reflect on the current situation and consider who the interlocutor is before selecting the most appropriate skill. This internal cognitive process is often represented as an explanation/ra-

tionale (Zhou et al., 2021; Wei et al., 2022). These explanations and skill choices are significantly influenced by the characteristics of the interlocutor, such as demographics, persona, and relationship.

2.2 Why "Skill-of-Mind" is necessary?

At the heart of the conversation is social interaction (Myllyniemi, 1986), a domain where current LLMs have limited understanding and struggle to effectively handle social interactive scenarios (Zhou et al., 2023; Liu et al., 2023). As a result, generating more engaging and natural responses directly through LLM-based conversational agents is challenging. This is because LLMs are primarily designed to solve complex reasoning tasks as general agents through alignment tuning (Ouyang et al., 2022; Chung et al., 2024), making them ill-suited to function as social dialogue agents. By introducing guidance based on the concept of *skill-of-mind*, LLM-based conversational agents can more effectively navigate social interactions. Current LLMs demonstrate better alignment and are capable of following user queries, so grounding responses in *skill-of-mind* can help narrow down the response options and focus on skill-specific aspects. This leads to more accurate outputs by reducing the range of possible responses (i.e., one-to-many problem). In practical applications, most conversational systems adopt a modular approach (Lee et al., 2023) where multiple skill-specialized dialogue agents are used to strengthen user interaction. In these systems, several skill-specific agents generate responses simultaneously, and a re-ranking module (Bae et al., 2022b) selects the best one for the given dialogue situation. While this approach is effective, it is also resource-intensive. In real-world scenarios, where multiple conversational skills are required, even with parallel processing, inference may not be as fast as using a single skill-expert agent. By focusing on a single agent grounded in *skill-of-mind*, we can potentially improve inference speed and reduce latency.

2.3 Ingredients for "Skill-of-Mind"

The concept of "skill-of-mind" consists of three main components: (1) social context, (2) explanation, and (3) conversational skill, which are described as follows.

Social Context Information. Socially interactive dialogues involve a wide range of social dynamics, such as demographics, personal experiences, and relationships. We believe that these factors influence the *skill-of-mind*. For instance, emotional empathy is more appropriate in conversations with a significant romantic partner than with an AI teaching mentor. Therefore, when considering social context information, we take into account various elements, such as the current situation or narrative, social relationships, persona, memory, and past dialogue histories. [2]

Explanation/Rationale. This involves interpreting and understanding the current situation to determine the most optimized conversational skill for generating an engaging response that strengthens social rapport (Zech and Rimé, 2005) with the interlocutor in the given dialogue. To achieve a higher quality of explanation, similar to how a human would respond, we adopt *perspective-taking*-style (Davis, 1983; Ruby and Decety, 2004; Kim et al., 2021), which prompts GPT-4 to imagine the actual speaker in the dialogue. The explanation is represented in free-form sentences.

Conversational Skill. As discussed in § 2.1, there is a broad range of conversational skills in real-world scenarios, such as empathy or persona management in chitchat, hotel reservations in task-oriented dialogues, and memory recall in long-term conversations. To account for this spectrum, we design a taxonomy that covers *fine-grained* levels of conversational skills across diverse scenarios. We develop a hierarchical taxonomy that encompasses distinct and *non-overlapping* [3] categories of skills. At the first level, we identify five main categories: (1) Interpersonal Skills, (2) Memory & Knowledge Management Skills, (3) Cognitive & Problem-Solving Skills, (4) Communication & Listening Skills, and (5) Task-Oriented Skills.

- **Interpersonal Skills:** These skills are essential for enhancing social interaction by requiring a deep understanding of the interlocutor's emotional state and adapting to their personality or relationship dynamics for more seamless and engaging communication. They also involve demonstrating prosocial behavior

[2] Note that we do not generate this information from scratch; rather, it originates from the source dialogue dataset used in this work.

[3] In reality, it is not always necessary to assign each conversational skill to a single category since many skills share overlapping characteristics. For instance, the skill of "Active Listening" can also be categorized under "Interpersonal Skills." However, for clarity in this work, we structure the taxonomy with *non-overlapping* conversational skill categories.

in problematic situations. We also consider image-sharing behavior, which frequently occurs via instant messaging tools. This category includes Empathy, Personal Background, Persona Recall, Self-Disclosure, Negotiation, Conflict Resolution, Conflict Avoidance, Persuasion, Commonsense Understanding, Cultural Sensitivity, Ethics, Harmlessness, Avoiding Social Bias, Helpfulness, Mentoring, Image Commenting, and Image Sharing.

- **Memory & Knowledge Management Skills:** These skills are primarily used to provide knowledgeable responses by sharing or acquiring information and recalling memories, which is important for maintaining long-term communication, particularly in senior care services (Bae et al., 2022a,b). This category includes Memory Recall, Knowledge Sharing, Knowledge Acquisition, and Knowledge Searching.

- **Cognitive & Problem-Solving Skills:** These skills are required for solving complex problems or performing factual reasoning tasks. This category includes Critical Thinking, Logical Thinking, Creative Problem Solving, Factual Problem Solving, and Decision-Making.

- **Communication & Listening Skills:** Effective listening is critical in the communication process (Main, 1985; Castleberry and Shepherd, 1993). Therefore, we include these skills in our taxonomy, which encompasses Clarification, Confirmation, Rephrasing, Echoing, Topic Transition, Rhetoric, Active Listening, Reflective Listening, and Immediate Response.

- **Task-Oriented Skills:** In practical scenarios, humans often request conversational agents (e.g., Alexa [4]) to perform tasks such as hotel or restaurant reservations, provide weather information, or offer movie recommendations. We also consider these skills, which include Recommendation, Task Execution, and Urgency Recognition.

2.4 Dataset Construction Process

Based on the above ingredients of *skill-of-mind* (§ 2.3), we first collect source dialogue datasets and annotate them with *skill-of-mind*, resulting in MULTIFACETED SKILL-OF-MIND.

[4] https://developer.amazon.com/en-US/alexa

Step 1: Source Dataset Collection. To build more flexible and versatile *skill-of-mind*-infused LLM, as a source data, we collect 12 multi-turn dialogue datasets, which are publicly available online: SODA (Kim et al., 2022a), CONVERSATIONCHRONICLES (Jang et al., 2023), PROSOCIALDIALOGUE (Kim et al., 2022b), EMPATHETICDIALOGUES (Rashkin, 2018), Wizard of Wikipedia (Dinan et al., 2018), CACTUS (Lee et al., 2024c), CASINO (Chawla et al., 2021), MultiWOZ 2.2 (Zang et al., 2020), PERSUASIONFORGOOD (Wang et al., 2019), PEARL (Kim et al., 2024), SYNPERSONACHAT (Jandaghi et al., 2023), and STARK (Lee et al., 2024d). In total, we collect source dialogues from the training sets. We then split each dialogue into sub-dialogues by focusing on one-sided exchanges. For example, given a dialogue $\mathcal{D} = \{(s_i, u_i)\}_{i=1}^{4}$, we create two sub-dialogues: $\mathcal{D}_1 = \{(s_i, u_i)\}_{i=1}^{2}$ and $\mathcal{D}_2 = \{(s_i, u_i)\}_{i=1}^{4}$. We remove sub-dialogues with fewer than four turns, as we believe that early in the dialogue, there is a higher distribution of non-informative skills, such as greetings, rather than informative skills. We then randomly sample sub-dialogues from each source dataset in specific proportions. As a result, we obtain a total of 100K dialogues.

Step 2: Annotating *Skill-of-Mind*. We prompt GPT-4 (Achiam et al., 2023) (i.e., `gpt-4-turbo`) to annotate *skill-of-mind* into the collected source dialogues. Specifically, it provides internal reasoning about which skills are appropriate for the next turn response in the dialogue and identifies the relevant multifaceted conversational skills from the predefined taxonomy (§ 2.3), taking into account the interlocutor's perspective (i.e., perspective-taking). Each instance in MULTIFACETED SKILL-OF-MIND consists of three input components (social context information, dialogue, next response) and two output components (explanation, skill).

The input components are described as follows:

- **Dialogue:** A dialogue between two speakers from the collected source datasets in step (1).

- **Next Response:** The next response in the dialogue, which should align with the relevant explanation and conversational skill. Given the subjective nature of dialogue, if only the dialogue is provided without a golden response, GPT-4 can still generate plausible explanations and skills that are not compatible with

Dataset	Train?	Explanation?	# of D.	# of S.
BST (Smith, 2020)	✓	✗	6,808	3
BS𝔹T (Kim et al., 2022c)	✓	✗	300,000	3
FLASK (Ye et al., 2023)	✗	✗	1,740	12
MULTIFACETED SKILL-OF-MIND	✓	✓	99,997	38+

Table 1: Comparison of MULTIFACETED SKILL-OF-MIND with existing datasets regarding skills: Blended-SkillTalk (BST), Blended Skill BotsTalk (BS𝔹T), and FLASK. D. and S. denote the dialogue and skill, respectively. The "+" next to 38 indicates the potential for additional skills as GPT-4 sometimes generates new skills not present in the predefined collection (e.g., *feedback giving*). To build a more flexible model, we include these cases in our training dataset.

the natural flow of the original dialogue.

- **Social Context Information:** Social context information encompasses various social dynamics, which vary depending on the source dialogues. For example, this includes social narratives in SODA and demographic factors (e.g., age, gender, birthplace, residence), personal narratives, or past session dialogue summaries in STARK.

The output components are described as follows:

- **Explanation:** A rationale explaining which skill is necessary to maintain continuous interaction with the interlocutor, given the input dialogue and the next response. To create more realistic explanations, GPT-4 is induced to engage in a perspective-taking process.

- **Conversational Skill:** Based on the explanation, one or more conversational skills relevant to the next response are selected from the predefined skill collections.

The prompt template is presented in Appendix E.We instruct GPT-4 to produce a structured output in JSON format, excluding any cases that fail to parse correctly. In total, we obtain 99,997 annotations (approximately 100K) and split the data into a 9:1 train/test ratio for evaluation. In instances with multiple *skill-of-mind* annotations, we randomly select one for training THANOS . An example of *skill-of-mind* annotation is presented in Table 2.

2.5 Analysis

Comparison to Existing Datasets. In Table 1, we compare MULTIFACETED SKILL-OF-MIND with other existing datasets that include certain skills. In summary, MULTIFACETED SKILL-OF-MIND is the first dataset to contain both explanations and skills. Although the number of dialogues is smaller than that of the BS𝔹T dataset, we include a greater number of conversational skills, which enhances the generalizability of the trained model. Compared to FLASK, our dataset also includes a larger variety of skills, whereas FLASKis designed to evaluate fine-grained LLM capabilities and primarily focuses on instruction-based skills. In contrast, our dataset offers a comparable dialogue size, and a substantial number of skills, and includes both explanations and skills, making it a robust resource for generalizable skill prediction.

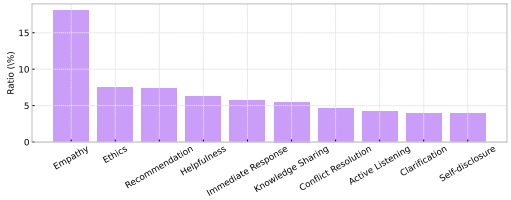

Figure 2: The ration (%) of Top-10 conversational skill categories in MULTIFACETED SKILL-OF-MIND.

Distribution of *Skill-of-Mind*. Figure 2 shows the distribution of the Top-10 conversational skill categories generated by GPT-4 (as discussed in § 2.4). We analyzed a total of 109,591 *skill-of-mind* annotations (considering multiple annotations per dialogue). The most prominent skill is Empathy, likely due to the presence of socially interactive datasets, such as SODA, CONVERSATIONCHRONICLES, and EMPATHETICDIALOGUES, which contain a large proportion of empathetic interactions — crucial in social dialogues. Additionally, Ethics and Helpfulness also occupy significant portions, indicating that the model trained on our dataset may serve as a *safeguard* to promote prosocial behavior.

Human Evaluation. To assess the quality of MULTIFACETED SKILL-OF-MIND, we conduct a human evaluation based on five criteria: (1) relevance, (2) plausibility, (3) understanding, (4) skill alignment, and (5) skill adequacy. Each human evaluator rates 100 randomly sampled instances on a 4-point Likert scale for all criteria. Criteria (1-3) measure the quality of the *explanation*, while criteria (4-5) assess the quality of the *conversational skill*. Further details about each evaluation criterion and the recruitment process for human evaluators are provided in the Appendix B and Appendix D.

Social Context Information:
Sawyer tells Gregg that he needs help with groceries tonight since he has to cook dinner for everyone. They go together and purchase the items on the list including chicken, broccoli, and pasta.

Dialogue:
Sawyer: Hey Gregg, do you think you could help me out with groceries tonight? I've got to cook dinner for everyone and it's a bit of a hassle by myself.
Gregg: Sure thing, Sawyer. What do you need to get?
Sawyer: Well, let's see...we need chicken, broccoli, and pasta. Oh, and some bread too.
Gregg: Alright, sounds like we can make a pretty good meal out of that. Let's head to the store then.
Sawyer: Thanks, Gregg. I really appreciate it.
Gregg: So, Sawyer, how come you're cooking dinner tonight?
Sawyer: Oh, you know...just felt like it. Plus, I love to cook so it's not a big deal.
Gregg: That's really nice of you. I'm sure everyone will appreciate it.
Sawyer: Yeah, I hope so! I just want to make something that everyone will enjoy.

Skill-of-Mind:
- **Explanation**: In responding to Sawyer, I want to show that I've been attentively listening to his thoughts and appreciating the effort he is putting into cooking dinner. By affirming his efforts and expressing confidence that everyone will enjoy his cooking, I am validating his feelings and intentions, which is key in making him feel supported.
- **Conversational Skill**: Active Listening

Table 2: A sample from MULTIFACETED SKILL-OF-MIND.

On average, we achieve notably high scores: 3.72 for relevance, 3.75 for plausibility, 3.74 for understanding, 3.64 for skill alignment, and 3.59 for skill adequacy. Additionally, we compute inter-rater agreement (IA) using Krippendorff's α, yielding a value of 0.62, which indicates a substantial level of agreement. These results demonstrate the reliability and quality of MULTIFACETED SKILL-OF-MIND, particularly with respect to generating human-like *skill-of-mind* in interactions.

3 THANOS : Skill-of-Mind-infused LLM

Backbone LLM. To enhance performance across various applications, we introduce a new family of skill-of-mind-infused LLMs with varying model sizes: THANOS -$\{1, 3, 8\}$B. For THANOS 1B, we fine-tune LLaMA-3.2-1B-Instruct; for THANOS 3B, we fine-tune LLaMA-3.2-3B-Instruct; and for THANOS 8B, we fine-tune LLaMA-3.1-8B-Instruct using MULTIFACETED SKILL-OF-MIND.

Input & Output. During THANOS training, we provide social context information and dialogue as input prompts. Since each source dialogue in our dataset contains varying levels of social context information, we design source-specific social context prompt templates. For each source dialogue, we create five different social context prompt templates, randomly sampling one during training for flexible generation. The social context prompt templates are presented in the Appendix A. For the output, THANOS is trained to sequentially generate an explanation, followed by conversational skill, similar to a Chain-of-Thought (Wei et al., 2022) fine-tuning approach. To mitigate degeneration issues, we introduce a `[RESULT SKILL]` token between the explanation and the conversational skill, as seen in prior work (Kim et al., 2023).

Implementation Details. We fine-tune THANOS-$\{1, 3, 8\}$B using LoRA (Hu et al., 2021) and PyTorch Fully-Sharded Data Parallel (FSDP). LoRA is applied to all linear layers with a rank of 256 and an alpha of 256. We set the maximum number of epochs to 3, with a batch size of 8 per GPU, using a StepLR scheduler and a learning rate of 1e-5 with the AdamW optimizer. All experiments are conducted on 8 NVIDIA A100 GPUs (40 GB).

4 Experiments

4.1 Experimental Setup

Task Definition. To evaluate the effectiveness of THANOS, we conduct three tasks: (1) *Skill Classification*: assessing the accuracy of skills generated by LLM-based agents and THANOS across various dialogue scenarios; (2) *Explanation Generation*: evaluating the quality of explanations produced in these scenarios; and (3) *Response Generation*: determining whether the generated skills and explanations improve the response quality in LLM-based agents.

Evaluation Datasets. We use different evaluation datasets for each of the three tasks: For (1) *Skill Classification*, we use Blended-SkillTalk (BST) (Smith, 2020), Blended Skill BotsTalk (BS$\mathbb{B}$T) (Kim et al., 2022c), MULTIFACETED SKILL-OF-MIND, and PhotoChat (Zang et al., 2021) to assess image-sharing behavior in multi-modal interactions, and PROSOCIALDIALOGUE (Kim et al., 2022b) for skills related to prosocial behavior, acting as a safeguard; For (2) *Explanation Generation*, we use MULTIFACETED SKILL-OF-MIND; For (3) *Response Generation*, we use out-of-domain dialogue datasets like DAILY-

DIALOG (Li et al., 2017), following previous methods (Kim et al., 2022a; Chae et al., 2023), and for in-domain settings, we use SODA and PROSOCIAL-DIALOGUES to evaluate how well THANOS promotes prosocial behavior.

Baselines. Since our goal is to enhance the quality and sophistication of responses generated by open-sourced LLM-based conversational agents through the incorporation of *skill-of-mind* capabilities, we select three different, widely-adopted LLM-based conversational agents as baselines: 1) `Gemma-2-2B`, 2) `LLaMA-3.1-8B` (which also serves as the backbone for our THANOS 8B), and 3) `COSMO-XL` (Kim et al., 2022a) which is specialized to social dialogue.

Evaluation Metrics. For (1), we measure accuracy (%). For (2), we evaluate using BLEU-1/2/4 (Papineni et al., 2002) and ROUGE-L (Lin, 2004). For (3), we apply the same metrics as in (2). However, due to the inherently subjective nature of dialogue evaluation, in addition to BLEU and ROUGE, we conduct a human evaluation based on four criteria: (1) naturalness, (2) engagingness, (3) consistency, and (4) overall quality.

4.2 Results

THANOS effectively infers the *skill-of-mind* process. As shown in Table 3, we present the comparative performance of skill classification and explanation generation on the MULTIFACETED SKILL-OF-MIND test set. Overall, THANOS demonstrates significantly better performance on both tasks compared to other LLM-based conversational agents. These results suggest that existing LLM-based conversational agents have a limited understanding of interaction scenarios, leading to a substantially reduced ability to infer the *skill-of-mind*. In contrast, THANOS effectively simulates the *skill-of-mind* process, similar to how humans do, benefiting from MULTIFACETED SKILL-OF-MIND. Furthermore, as the size of the backbone model increases, performance continues to improve.

THANOS demonstrates strong generalizability. In Table 3, THANOS performs well in out-of-domain settings. Compared to baselines, THANOS outperforms on the BST, which was not used during its training. Additionally, on the PROSOCIAL-DIALOGUE, our model achieves significant performance, indicating its potential for safety detection. Furthermore, THANOS shows comparable image-sharing capabilities on the PhotoChat. These results suggest that THANOS has strong generalization performance.

THANOS enhances the quality of the generated responses. Table 4 presents the performance of the response generation task for LLM-based conversational agents with and without THANOS. Overall, THANOS significantly increases the quality of generated responses, suggesting the effectiveness of *skill-of-mind* as socially-aware guidance. Scaling up THANOS improves performance further, though the efficient 1B-size version also achieves notable improvements, indicating the potential for use in mobile environments. Surprisingly, THANOS significantly boosts the performance of `Gemma-2-2B`, demonstrating that *skill-of-mind* is effectively compatible with efficient LLM-based conversational agents without compromising generalizability. In addition, in the PROSOCIALDIALOGUE setting, THANOS significantly increases performance by a large margin, suggesting that *skill-of-mind* successfully induces prosocial behavior in LLM-based agents, serving as a safeguard. This highlights its potential as a safety mechanism (Han et al., 2024). Furthermore, even in social conversational agents like `COSMO-XL`, THANOS enables the generation of more adaptive and higher-quality responses, implying that *skill-of-mind* is fully compatible with the socially aligned foundation model.

THANOS enhances more human-friendly responses. Table 5 provides a detailed analysis of the effect of THANOS on two datasets. THANOS induces agents to generate more human-like, empathetic responses, as evidenced by improvements in the accuracy of empathetic intent classification and emotion classification. Additionally, the diff-Ex score shows a small difference between the golden human response and the predicted response. In the PROSOCIALDIALOGUEdataset, the frequency of the "casual" label increases, while the "caution" label decreases. These results suggest that THANOS helps LLM-based agents exhibit more human-like and prosocial behavior.

Results of head-to-head evaluation. Table 6 shows the human evaluation results on the DAILY-DIALOG dataset based on five evaluation criteria: (1) naturalness, (2) specificity, (3) consistency, (4) engagingness, and (5) overall quality. For this, we randomly sampled 70 dialogues and asked human evaluators to choose the better response between

Models	Skill Classification					Explanation Generation				
	Ours	Prosocial	BST	PhotoChat	Avg.	B-1	B-2	B-4	R-L	BertScore
Gemma-2-2B	3.30	12.70	1.97	5.06	5.76	8.00	2.40	0.30	4.30	29.33
LLaMA-3.1-8B	5.20	18.42	7.10	4.03	8.69	10.90	3.90	0.70	3.80	26.18
THANOS 1B	27.50	50.40	24.16	12.60	28.67	29.80	14.10	4.20	18.70	88.49
THANOS 3B	28.80	50.80	**24.85**	14.87	29.83	30.90	14.70	4.90	19.50	88.45
THANOS 8B	**29.70**	**53.80**	23.08	**15.81**	**30.60**	**31.20**	**15.10**	**5.40**	**20.10**	**88.53**

Table 3: Comparison of skill classification accuracy (%) and explanation generation on MULTIFACETED SKILL-OF-MIND (Ours), PROSOCIALDIALOGUE (Prosocial), BST, and PhotoChat datasets. B-1/2/4 refer to BLEU-1/2/4 (Papineni et al., 2002), and R-L refers to ROUGE-L (Lin, 2004) for simplicity.

Models	DAILYDIALOG					EMPATHETICDIALOGUES					PROSOCIALDIALOGUE				
	B-1	B-2	B-4	R-L	BertScore	B-1	B-2	B-4	R-L	BertScore	B-1	B-2	B-4	R-L	BertScore
Gemma-2-2B	6.02	2.28	0.57	11.53	82.71	5.75	1.76	0.37	10.12	84.03	13.77	4.32	0.76	12.25	85.53
+ THANOS 1B	10.55	3.66	0.87	11.66	83.86	13.37	**4.1**	**1.01**	11.38	85.47	18.48	5.66	0.95	12.63	86.2
+ THANOS 3B	11.01	3.63	0.82	11.74	83.72	13.68	4.03	0.87	11.56	85.52	18.51	5.8	**1.07**	**12.98**	86.25
+ THANOS 8B	**11.16**	**3.94**	**0.97**	**12.46**	**83.82**	**13.67**	4.01	0.76	**11.76**	**85.54**	**18.54**	**5.84**	1.01	12.8	**86.26**
LLaMA-3.1-8B	**11.6**	**4.44**	**1.16**	**11.5**	83.51	9.74	3.02	0.6	10.42	84.73	15.12	5.49	1.24	13.07	85.93
+ THANOS 1B	9.93	3.49	0.85	10.82	83.38	10.63	**3.18**	0.55	10.65	84.88	**17.83**	**6.03**	**1.27**	13.17	86.01
+ THANOS 3B	9.95	3.42	0.79	10.97	83.42	10.94	3.27	**0.78**	**10.94**	**84.94**	17.44	5.89	1.22	13.13	86.02
+ THANOS 8B	10.95	4.09	1.06	11.45	**83.59**	**10.65**	2.9	0.5	10.27	84.87	17.72	5.86	1.16	**13.27**	**86.06**
COSMO-XL	3.92	0.64	0.12	3.32	38.02	**10.76**	2.7	0.29	7.19	70.46	8.35	2.76	0.61	10.2	72.92
+ THANOS 1B	10.15	2.78	0.33	6.95	67.64	10.31	**2.81**	0.32	8.35	**74.22**	13.71	4.49	0.91	10.73	73.1
+ THANOS 3B	10.6	2.58	0.31	6.45	66.25	10.15	2.71	**0.46**	**8.52**	73.44	**15.06**	**4.87**	0.96	11.01	73.23
+ THANOS 8B	**10.71**	**2.88**	**0.51**	**7.1**	**69.05**	10.24	2.7	0.37	8.49	74.11	14.55	4.71	**1**	**11.18**	**75.07**

Table 4: Automatic evaluation results of response generation task on DAILYDIALOG, EMPATHETICDIALOGUES, PROSOCIALDIALOGUE datasets.

	EMPATHETICDIALOGUES			PROSOCIALDIALOGUE		
	Intent	Emotion	diff-EX	Casual	Caution	Intervention
Gemma-2-2B	23.5	14.7	1.42	74.3	22.4	2.2
+ THANOS 1B	23.8	**15.4**	1.08	**88.0**	**10.1**	1.9
+ THANOS 3B	**25.7**	14.1	**1.06**	85.7	12.8	**1.5**
+ THANOS 8B	24.1	**15.4**	**1.06**	87.2	11.1	1.7
COSMO-XL	20.5	12.3	1.83	70.3	13.9	1.0
+ THANOS 1B	**20.6**	13.3	**1.45**	70.6	13.5	**0.9**
+ THANOS 3B	20.4	**13.9**	1.50	70.8	13.1	1.2
+ THANOS 8B	20.5	13.6	1.51	**74.1**	**12.4**	**0.9**

Table 5: Detailed performance comparison on empathy- and prosocial-related datasets. The metrics Intent, Emotion, and diff-EX are designed to evaluate sophisticated aspects of empathetic responses. We refer to previous work (Lee et al., 2022a) for details on the evaluation process. A lower diff-EX value indicates a more human-like response. For a detailed analysis of prosocial behavior, we measure the ratio of safety labels using the Canary model (Kim et al., 2022b), a safety classification model. If the sum of the safety label ratios does not equal 100, it indicates degeneration has occurred.

	Natural	Specific	Consistent	Engaging	Overall
Gemma-2-2B	38.6	**50**	47.2	44.3	42.9
+ THANOS 8B	**61.4**	**50**	**52.8**	**55.7**	**57.1**
LLaMA-3.1-8B	28.6	**54.3**	41.4	42.9	41.4
+ THANOS 8B	**71.4**	45.7	**58.6**	**57.1**	**58.6**

Table 6: Head-to-head evaluation between LLM-based agents and those equipped with THANOS 8B on response generation for the DAILYDIALOG dataset.

the LLM and the LLM with THANOS. Overall, THANOS effectively helps LLM-based agents generate responses that are more preferred by real humans. However, in terms of specificity, THANOS does not help the LLM agent achieve better performance. These results suggest that current LLM-based agents are mainly trained to provide helpful and informative responses to users' complex queries, thereby yielding better specificity. Future work should focus on building conversational agents possessing both social reasoning and complex reasoning.

5 Related Work

Conversational Skills. There have been a few studies that cover conversational skills. For example, BlendedSkillTalk (Smith, 2020) was the first to propose a dialogue dataset encompassing multiple conversational skills, including *persona*, *empathy*, and *knowledge*. Blended Skill BotsTalk (Kim et al., 2022c) also addresses the same conversational skills as BlendedSkillTalkbut scales up the dataset size through an automatic dataset construction method. Unlike these two datasets, FLASK (Ye et al., 2023) focuses on fine-grained skills for evaluating the multi-capabilities of instruction-aware LLMs, though it is not used for training purposes. In contrast, our work introduces the concept of *skill-of-mind* and presents MULTIFACETED SKILL-OF-MIND, where each dialogue includes both an *explanation* and a *conversational skill*. Compared to other datasets, MULTIFACETED SKILL-OF-MIND incorporates *explanation*, which is grounded in a perspective-taking approach, and covers a larger number of conversational skills.

LLM-based Conversational Agents. Recent LLM-based conversational agents, such as ChatGPT (OpenAI, 2023), GPT-4 (Achiam et al., 2023), and LLaMA-3 (AI@Meta, 2024), are built on top of large pre-trained LLMs via instruction fine-tuning or RLHF. These agents have been widely used to construct socially-aware dialogue datasets, such as SODA(Kim et al., 2022a) and STARK(Lee et al., 2024d), through symbolic knowledge distillation frameworks. However, as previous studies have reported (Zhou et al., 2023), these agents still show limited performance in socially interactive scenarios, especially in the case of open-source models where performance tends to degrade more than in closed-source models. We argue that, for open-source agents, directly generating socially-aware responses poses a greater challenge. To alleviate this burden, we propose that guiding response generation using the *skill-of-mind* concept can enhance the performance of open-source agents, particularly in terms of the quality of their responses.

6 Conclusion

In this work, we introduce the concept of *skill-of-mind* that involves interpreting social contexts and selecting appropriate conversational skills. We also present MULTIFACETED SKILL-OF-MIND, a multi-turn dataset annotated with diverse *skill-of-mind* dynamics, and propose THANOS , a family of *skill-of-mind*-infused LLMs, demonstrating their effectiveness across various tasks. Our work highlights the potential to enhance socially aware conversations in open-source models through skill-based guidance, paving the way for future advancements in skill-driven conversational AI.

Limitations

Extending the Generalizability of *Skill-of-Mind*. Although THANOS is trained on MULTIFACETED SKILL-OF-MIND, which includes multifaceted *skill-of-mind* across diverse dialogue scenarios (e.g., counseling, task-oriented interactions), this work focuses on enhancing LLM-based conversational agents (i.e., `LLaMA-3.1-8B`, `Gemma-2-2B`) to generate more engaging and natural responses based on the *skill-of-mind* guidance provided by THANOS . To further verify the extensive *generalization* capabilities of THANOS , we need to conduct additional experiments in more varied dialogue scenarios (Zhang et al., 2023; Kim et al., 2024; Lee et al., 2024c). For instance, THANOS could be beneficial for psychological counseling services or adaptable to off-the-shelf home assistants (e.g., `Alexa`). We leave this for future work.

Building a *Skill-of-Mind*-Embedded Dialogue Agent. In this work, we build a *skill-of-mind*-infused LLM, THANOS , and demonstrate that incorporating *skill-of-mind* enhances the generation of more natural, socially aware responses in LLM-based conversational agents. However, the current approach still relies on providing *skill-of-mind* through the LLM's input prompt, which means the core of the LLM-based agent still lacks the inherent ability to fully comprehend social interactions (Zhou et al., 2023). Inspired by the recent success of knowledge-embedded, task-specific foundation models (Lee et al., 2024a; Yoon et al., 2024), we need to build a more advanced *skill-of-mind*-infused dialogue agent by embedding *skill-of-mind* directly into the model.

Ethical Considerations

In constructing MULTIFACETED SKILL-OF-MIND, we use the PROSOCIALDIALOGUE dataset as the source dialogue. Although this dataset focuses on promoting *prosocial behavior*, some instances may contain relatively unsuitable phrases (e.g., politics). Consequently, THANOS trained on MULTIFACETED SKILL-OF-MIND could be exposed to

these harmful instances. However, the goal of this work is to generate *skill-of-mind* in various dialogue situations, including those involving *prosocial behavior*, rather than generating harmful or offensive responses. Nonetheless, it is important to use our model cautiously and with care to avoid unintended consequences.

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A Prompt Template for Social Context Information

Table 7, Table 8, Table 9, Table 10, Table 11, Table 12, Table 13, Table 14, Table 15, Table 16 show social context template for PROSOCIALDIALOGUE (Kim et al., 2022b), STARK (Lee et al., 2024d) (first round session), STARK (Lee et al., 2024d) (N-th round session), CACTUS (Lee et al., 2024c), SYN-PERSONACHAT (Jandaghi et al., 2023), CASINO (Chawla et al., 2021) (sentence format), CASINO (Chawla et al., 2021) (structured format), PEARL (Kim et al., 2024), PERSUASIONFORGOOD (Wang et al., 2019), EMPATHETICDIALOGUES (Rashkin, 2018).

B Human Evaluation Questionnaire

This section presents the list of questions and multiple-choice options used for the human ratings represented in Section 2.

B.1 Human Ratings

- **Relevance:** How relevant is the given explanation to the current dialogue situation and the social context?

  Options: 1: Not at all / 2: A little / 3: Somewhat / 4: A lot

- **Plausibility:** Does the given explanation seem plausible, as if a human would think in a real-world scenario?

  Options: 1: Not at all / 2: A little / 3: Somewhat / 4: A lot

- **Understanding:** How does the given explanation demonstrate understanding of the current dialogue situation and the social context?

  Options: 1: Not at all / 2: A little / 3: Somewhat / 4: A lot

- **Skill Alignment:** Does the selected conversational skill align well with the provided explanation?

  Options: 1: Not at all / 2: A little / 3: Somewhat / 4: A lot

- **Skill Adequacy:** Do the conversational skills currently used seem appropriate for generating a suitable response in the upcoming turn?

  Options: 1: Not at all / 2: A little / 3: Somewhat / 4: A lot

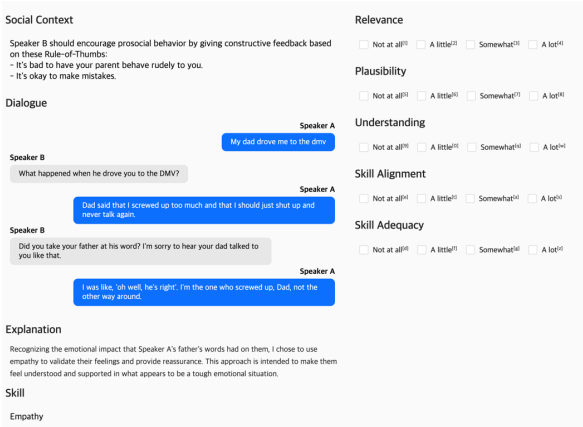

Figure 3: A screenshot of human rating evaluation for MULTIFACETED SKILL-OF-MIND.

B.2 Head-to-Head Comparison

- **Naturalness:** Which response is more natural?

  Options: Definitely A / Slightly A / Slightly B / Definitely B

- **Consistent:** Which response is more consistent?

  Options: Definitely A / Slightly A / Slightly B / Definitely B

- **Specificity:** Which response is more specific?

  Options: Definitely A / Slightly A / Slightly B / Definitely B

- **Engagingness:** Which response is more engaging?

  Options: Definitely A / Slightly A / Slightly B / Definitely B

- **Overall:** Which response do you like more overall?

  Options: Definitely A / Slightly A / Slightly B / Definitely B

C Human Evaluation System

We show a screenshot of the human evaluation system in Figure 3 and Figure 4. We implement this system using Label Studio (Tkachenko et al., 2020-2022).

D Details of Human Evaluation

We recruited 15 individuals, unknown to us, who are either graduate or undergraduate students. Prior

Template for Social Context Information in PROSOCIALDIALOGUE
Speaker B should foster prosocial behavior by providing constructive feedback based on these Rule-of-Thumbs:\n- {rots}
Speaker B should encourage prosocial behavior by giving constructive feedback based on these Rule-of-Thumbs:\n- {rots}
To promote positive behavior, Speaker B should offer constructive feedback following these Rule-of-Thumbs:\n- {rots}
Guided by these Rule-of-Thumbs, Speaker B should encourage prosocial behavior through constructive feedback:\n- {rots}
Speaker B is expected to provide constructive feedback to encourage positive interactions, using these Rule-of-Thumbs:\n- {rots}

Table 7: Template for social context information in PROSOCIALDIALOGUE (Kim et al., 2022b). {rots} denotes Rule-of-Thumbs (RoTs).

Template for Social Context Information in STARK (First Round Session)
{name} is {age} years old, born in {birthplace}, and currently lives in {residence}. {event}
{name}, aged {age}, was born in {birthplace} and resides in {residence}. {event}
{name}, who is {age}, was born in {birthplace} and now lives in {residence}. {event}
{name} is {age}, originally from {birthplace}, and now living in {residence}. {event}
{name} is {age} years old, born in {birthplace}, and resides in {residence}. {event}

Table 8: Template for social context information in STARK (Lee et al., 2024d) (first round session).

Template for Social Context Information in STARK (N-th Round Session)
{name} is {age} years old, born in {birthplace}, and currently lives in {residence}. After {time_interval}, {name} has gone through {experience}, and now {event}
{name}, aged {age}, was born in {birthplace} and now resides in {residence}. Following {time_interval}, {name} experienced {experience}, and {event}
{name}, who is {age} years old, originally from {birthplace} and living in {residence}, went through {experience} after {time_interval}, and now {event}
{name} is {age}, born in {birthplace}, and currently resides in {residence}. After {time_interval} of {experience}, {name} has now {event}
{name}, {age} years old, from {birthplace} and residing in {residence}, has experienced {experience} over {time_interval}, and as a result, {event}

Table 9: Template for social context information in STARK (Lee et al., 2024d) (N-th round session).

Template for Social Context Information in CACTUS
Client's attitude is {client attitude}. The client's intake form is as follows:\n{client intake form}.
The client has an attitude of {client attitude}. Below is the client's intake form:\n{client intake form}.
With an attitude of {client attitude}, the client's intake form details are:\n{client intake form}.
Client's attitude: {client attitude}. Intake form information:\n{client intake form}.
The client's attitude is {client attitude}. Here is their intake form:\n{client intake form}.

Table 10: Template for social context information in CACTUS (Lee et al., 2024c).

Template for Social Context Information in SYN-PERSONACHAT
User 1's Persona Information:\n- {user1 persona}\n\nUser 2's Persona Information:\n- {user2 persona}
User 1's Profile:\n- {user1 persona}\n\nUser 2's Profile:\n- {user2 persona}
Details of User 1's Persona:\n- {user1 persona}\n\nDetails of User 2's Persona:\n- {user2 persona}
Persona for User 1:\n- {user1 persona}\n\nPersona for User 2:\n- {user2 persona}
Information about User 1's Persona:\n- {user1 persona}\n\nInformation about User 2's Persona:\n- {user2 persona}

Table 11: Template for social context information in SYN-PERSONACHAT (Jandaghi et al., 2023).

Template for social context information in CASINO (sentence format)
Speaker A is a {speaker_a_age}-year-old {speaker_a_ethnicity} {speaker_a_gender} who has a {speaker_a_education} education. Their social value orientation is {speaker_a_svo}. According to the Big Five personality traits, they score {speaker_a_extraversion} in extraversion, {speaker_a_agreeableness} in agreeableness, {speaker_a_conscientiousness} in conscientiousness, {speaker_a_emotional_stability} in emotional stability, and {speaker_a_openness_to_experiences} in openness to experiences. In the negotiation, Speaker A's highest priority is {speaker_a_value2issue_high}, for which they reasoned: "{speaker_a_value2reason_high}". Their medium priority is {speaker_a_value2issue_medium}, with the reasoning: "{speaker_a_value2reason_medium}". Their lowest priority is {speaker_a_value2issue_low}, and they stated: "{speaker_a_value2reason_low}". — Speaker B is a {speaker_b_age}-year-old {speaker_b_ethnicity} {speaker_b_gender} who has a {speaker_b_education} education. Their social value orientation is {speaker_b_svo}. Their Big Five personality traits scores are {speaker_b_extraversion} in extraversion, {speaker_b_agreeableness} in agreeableness, {speaker_b_conscientiousness} in conscientiousness, {speaker_b_emotional_stability} in emotional stability, and {speaker_b_openness_to_experiences} in openness to experiences. During the negotiation, Speaker B's top priority is {speaker_b_value2issue_high}, and they explained: "{speaker_b_value2reason_high}". Their medium priority is {speaker_b_value2issue_medium}, with the reason: "{speaker_b_value2reason_medium}". Their lowest priority is {speaker_b_value2issue_low}, about which they mentioned: "{speaker_b_value2reason_low}".

Table 12: Template for social context information in CASINO (Chawla et al., 2021) (sentence format).

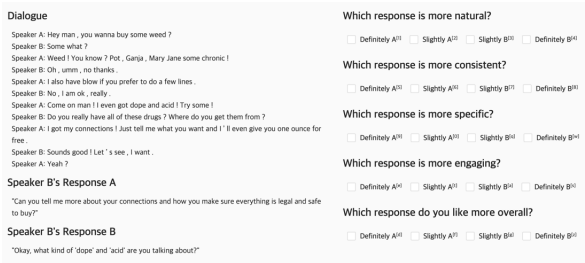

Figure 4: A screenshot of head-to-head comparison evaluation for DailyDialog (Li et al., 2017)

to participating in the experiment, they were provided with comprehensive instruction on the task, an overview of the *skill-of-mind*-annotated dialogue dataset, and a detailed explanation of the evaluation criteria. This preparatory phase lasted approximately roughly 15 minutes.

Template for social context information in CASINO (structured format)
Speaker A's Demographic Information:
- Age: {speaker_a_age}
- Gender: {speaker_a_gender}
- Ethnicity: {speaker_a_ethnicity}
- Education: {speaker_a_education}
Speaker A's Personality Information:
- Social Value Orientation (SVO): {speaker_a_svo}
- Big Five Personality Traits:
- Extraversion: {speaker_a_extraversion}
- Agreeableness: {speaker_a_agreeableness}
- Conscientiousness: {speaker_a_conscientiousness}
- Emotional Stability: {speaker_a_emotional_stability}
- Openness to Experiences: {speaker_a_openness_to_experiences}
Speaker A's Negotiation Information:
- Priority Order (value2issue):
- High: {speaker_a_value2issue_high}
- Medium: {speaker_a_value2issue_medium}
- Low: {speaker_a_value2issue_low}
- Personal Arguments (value2reason):
- High: {speaker_a_value2reason_high}
- Medium: {speaker_a_value2reason_medium}
- Low: {speaker_a_value2reason_low}
—
Speaker B's Demographic Information:
- Age: {speaker_b_age}
- Gender: {speaker_b_gender}
- Ethnicity: {speaker_b_ethnicity}
- Education: {speaker_b_education}
Speaker B's Personality Information:
- Social Value Orientation (SVO): {speaker_b_svo}
- Big Five Personality Traits:
- Extraversion: {speaker_b_extraversion}
- Agreeableness: {speaker_b_agreeableness}
- Conscientiousness: {speaker_b_conscientiousness}
- Emotional Stability: {speaker_b_emotional_stability}
- Openness to Experiences: {speaker_b_openness_to_experiences}
Speaker B's Negotiation Information:
- Priority Order (value2issue):
- High: {speaker_b_value2issue_high}
- Medium: {speaker_b_value2issue_medium}
- Low: {speaker_b_value2issue_low}
- Personal Arguments (value2reason):
- High: {speaker_b_value2reason_high}
- Medium: {speaker_b_value2reason_medium}
- Low: {speaker_b_value2reason_low}

Table 13: Template for social context information in CASINO (Chawla et al., 2021) (structured format).

Template for Social Context Information in PEARL
Seeker's overall movie preferences are represented as follows:\n{user persona}
Here is the seeker's complete movie profile:\n{user persona}
The seeker's general movie state is described below:\n{user persona}
Representation of seeker's overall movie interests:\n{user persona}
Below is the seeker's overall movie persona:\n{user persona}

Table 14: Template for social context information in PEARL (Kim et al., 2024).

Template for Social Context Information in PERSUASIONFORGOOD
Speaker A is attempting to persuade Speaker B.
In this scenario, Speaker A is the Persuader and Speaker B is the Persuadee.
Speaker A acts as Persuader, while Speaker B plays the role of Persuadee.
In the conversation, Speaker A is persuading Speaker B.
Speaker A aims to convince Speaker B.

Table 15: Template for social context information in PERSUASIONFORGOOD (Wang et al., 2019).

Template for Social Context Information in EMPATHETICDIALOGUES
Speaker A is feeling {emotion} because {situation}.
Due to {situation}, Speaker A's emotion is {emotion}.
Speaker A's emotional state: {emotion}; Situation: {situation}.
Because of {situation}, Speaker A is in a {emotion} mood.
The situation is {situation}, so Speaker A feels {emotion}.

Table 16: Template for social context information in EMPATHETICDIALOGUES (Rashkin, 2018).

E A Prompt Template for MULTIFACETED SKILL-OF-MIND

Prompt Template for *Skill-of-Mind* Generation

System Message:

You are a helpful assistant that generates the most appropriate conversational skill and corresponding explanation. Read the provided instruction carefully.

Instruction:

In the given dialogue, two speakers are communicating with each other, and each speaker has their own information such as demographics, preferences, persona, current situation/narrative, past dialogue summaries, episodic memory, or other relevant details. This information is represented in the "[Social Context]" part. In this dialogue, image-sharing moments sometimes occur, represented in the format of "[Sharing Image] <image_description>", where <image_description> represents the description of the shared image. You are also given the ideal response for the next turn in the given dialogue. Your task is to identify the most appropriate conversational skill that would lead to the ideal response in the given dialogue from the skill collection below, and explain why this particular skill was chosen. When generating the explanation, you should adopt the perspective of the speaker in the dialogue, selecting the skill based solely on the context of the given conversation. Do not consider the ideal response when generating your explanation; focus only on the given dialogue itself and why the chosen skill is the most suitable in that specific situation.

We provide the skill collection:
[Skill Collections]
- Empathy, Personal Background, Persona Recall, Self-disclosure, Negotiation, Conflict Resolution, Conflict Avoidance, Persuasion, Memory Recall, Topic Transition, Ethics, Harmlessness, Helpfulness, Avoiding Social Bias, Cultural Sensitivity, Commonsense Understanding, Rhetoric, Preference Elicitation, Knowledge Sharing, Knowledge Acquisition, Knowledge Searching, Active Listening, Factual Problem Solving, Logical Thinking, Critical Thinking, Creative Problem Solving, Immediate Response, Rephrasing, Echoing, Mentoring, Reflective Listening, Image-Sharing, Image-Commenting, Recommendation, Task Execution, Urgency Recognition, Clarification, Confirmation, Decision-making

Given the dialogue, social context information, and the next response, please brainstorm the most appropriate conversation skill and corresponding explanation.
[Social Context]
`{social_context}`

[Dialogue]
`{dialogue}`

[Next Response]
`{response}`

You should strictly follow the guidelines below:
[Guidelines]
- The answer should be represented in the form of a JSON list.
- Each entry in the list should be a Python dictionary containing the following keys: "skill", "explanation".

- The "skill" field should contain the one skill that is mostly required to generate the next response.
- The "explanation" field should provide a reason that occurs in the actual speaker's mind before selecting the skill, from the speaker's perspective.
- The "explanation" should be written from the perspective of the actual speaker who made the next response.
- You can choose one or multiple skills if necessary, but each skill must have its own explanation.

[Generated Skills and Explanations]